PopulationBalanceModels.jl

Define the physics once. Reuse the model.

Population mechanisms

Particles Cells Bubbles

Growth

Birth / Death

Aggregation / Breakage

Growth Rate Dispersion

Custom Terms

Shared forward model

Composable population balance

$$\frac{\partial n}{\partial t} = \sum_i f_i[n]$$

Select compatible numerics

Direct discretization

Moment formulations

Transient / Stationary

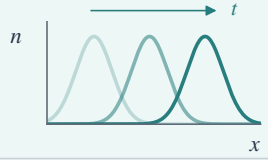
Optional coupling

Bulk balances · Reactor transport

Analysis

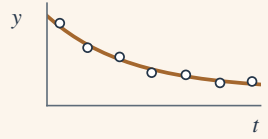
Simulate

Population dynamics



Estimate

Parameters from data



Quantify

Uncertainty MC / PCE

